

Customer Returns Analysis

Business Problem → Data Survey Schema → Needed Answers → Final Solution

1. Business Problem

Business question: Why are return rates high, and how can we reduce them?

The project analyzes customer-return survey data to identify the main reasons customers return products, understand what happens when customers want an exchange but the replacement is unavailable, and identify customers who could potentially be retained through a Smart Exchange approach.

Core questions

Area	What we need to understand
Return rate	How many customers returned and what is the overall return rate?
Return reasons	Which reasons are responsible for the largest share of returns?
Exchange availability	When customers want an exchange, why is the replacement unavailable?
Size / color	How much of the exchange problem is caused by unavailable desired size or color?
Smart Exchange	How many customers match the defined Smart Exchange candidate criteria?

2. Data Survey Schema Template

The final Gold customer-return table contains 145 unique records. The fields below are the schema used for the Power BI analysis.

Field	Purpose / meaning
response_id	Unique survey response identifier.
age_group	Age bucket: 18-25, 26-35, 36-45, 46-55, 56+.
gender	Customer gender category.
purchase_frequency	How frequently the customer purchases.
ever_returned	Whether the customer has ever returned: Yes / No.
return_count	Number of returns reported by the customer.
return_count_missing	Flag for originally missing return_count.
main_return_reason	Primary reason for the return.
wanted_exchange	Whether the customer wanted an exchange.
exchange_available	Whether an exchange was available.
replacement_unavailable	Reason the desired replacement was unavailable.
what_happened_when_unavailable	Outcome when replacement was unavailable.
ingested_at	Ingestion timestamp.
return_status	Gold-derived: Returned / Not Returned.
smart_exchange_candidate	Gold-derived Smart Exchange eligibility flag.
smart_exchange_type	Gold-derived type, e.g. Size Availability or Color Availability.

3. Needed Answers

The final report answers these questions in a business sequence.

Question	Finding
Total customers?	145 unique records.
Returned customers?	102.
Return rate?	70.34%.
Leading return reasons?	Wrong Size: 47; Wrong Color: 24; Poor Quality: 7; Damaged Product: 6.
Major replacement problem?	No Stock in Desired Size: 40; No Stock in Desired Color: 11.
Wanted exchanges?	72.
Smart Exchange candidates?	51.
Smart Exchange rate?	70.83% of wanted exchanges.
Candidate types?	Size Availability: 40; Color Availability: 11.
Strong root cause?	23 Wrong Size cases had no stock in the desired replacement size.

4. Final Solution

Proposed solution: Smart Exchange. Instead of immediately forcing a return when the desired replacement size or color is unavailable, the process attempts to identify a suitable exchange opportunity.

Solution workflow

Step	Action
1	Identify a customer who wants an exchange.
2	Check whether the requested replacement is unavailable.
3	Classify the opportunity, especially Size Availability or Color Availability.
4	Offer a Smart Exchange match when a suitable alternative exists.
5	Convert eligible return journeys into exchanges where possible.
6	Use Power BI to monitor return reasons and exchange opportunities.

Final business recommendation

Improve replacement inventory availability for popular sizes and colors, and introduce Smart Exchange before processing a return. Prioritize the strongest observed opportunity: Wrong Size / Wrong Color returns associated with unavailable desired replacement size or color.

Data limitation: The dataset does not provide a reliable historical return date, product category, geography, order value, refund amount, or revenue impact. The final dashboard therefore does not claim monthly trends, regional maps, revenue impact, or category return rates.

Power BI report structure

Page	Purpose
Page 1	Executive dashboard — KPIs, return reasons, exchange problems, Smart Exchange opportunity, and business insight.
Page 2	Root-cause deep dive — return reasons versus replacement-unavailable reasons.
Page 3	Smart Exchange concept — current experience vs proposed experience.
Page 4	Solution/process flow — current pain point, current vs proposed flow, and business benefits.